

Python String Methods

Coding Assignment — 36 Questions

Topics Covered	String methods — no loops, no conditionals, pure string operations
Difficulty Levels	10 Basic · 10 Easy · 10 Intermediate · 6 LeetCode-style
Total Questions	30 standard + 6 LeetCode-style = 36 questions

Section 1 — Basic

These 10 questions cover the most fundamental string methods. Students should be able to solve these immediately after the lesson.

Q1 Get the length of a string	
BASIC	<code>len()</code>
Given a string, find how many characters it contains including spaces.	
INPUT <code>s = "Hello, World!"</code>	EXPECTED OUTPUT <code>13</code>

Q2 Convert a string to uppercase	
BASIC	<code>.upper()</code>
Given a student's name in lowercase, convert it to all uppercase letters.	
INPUT <code>name = "sahil"</code>	EXPECTED OUTPUT <code>"SAHIL"</code>

Q3 Convert a string to lowercase	
BASIC	<code>.lower()</code>
A user typed their email in mixed case. Convert it to lowercase for storage.	
INPUT <code>email = "Sahil@GMAIL.COM"</code>	EXPECTED OUTPUT <code>"sahil@gmail.com"</code>

Q4 Remove whitespace from both ends

BASIC

`.strip()`

A form input came in with extra spaces around it. Clean it up.

INPUT

```
s = "  Python  "
```

EXPECTED OUTPUT

```
"Python"
```

Q5 Replace a word in a sentence

BASIC

`.replace()`

Replace the word 'bad' with 'good' in the given sentence.

INPUT

```
s = "Today was a bad day"
```

EXPECTED OUTPUT

```
"Today was a good day"
```

Q6 Check if a string starts with a word

BASIC

`.startswith()`

Check whether the sentence starts with the word 'Hello'.

INPUT

```
s = "Hello, how are you?"
```

EXPECTED OUTPUT

```
True
```

Q7 Check if a string ends with a suffix

BASIC

`.endswith()`

Check whether the filename ends with '.py'.

INPUT

```
filename = "assignment.py"
```

EXPECTED OUTPUT

```
True
```

Q8 Split a sentence into a list of words

BASIC

`.split()`

Split the sentence into individual words.

INPUT

```
s = "the cat sat on the mat"
```

EXPECTED OUTPUT

```
['the', 'cat', 'sat', 'on', 'the', 'mat']
```

--

Q9 Join a list of words into a sentence	
BASIC	<code>.join()</code>
Join the list of words into a single sentence separated by spaces.	
INPUT <code>words = ["Python", "is", "fun"]</code>	EXPECTED OUTPUT <code>"Python is fun"</code>

Q10 Find the position of a substring	
BASIC	<code>.find()</code>
Find the index where the word 'cat' first appears in the string.	
INPUT <code>s = "the cat sat on the mat"</code>	EXPECTED OUTPUT <code>4</code>

Section 2 — Easy

These 10 questions introduce slightly less obvious methods and require students to think about which method is the right tool.

Q11 Count how many times a letter appears	
EASY	<code>.count()</code>
Count how many times the letter 'a' appears in the string.	
INPUT <code>s = "banana"</code>	EXPECTED OUTPUT <code>3</code>

Q12 Capitalise the first letter	
EASY	<code>.capitalize()</code>
A student's name was entered in all lowercase. Capitalise only the first letter.	
INPUT <code>name = "alice"</code>	EXPECTED OUTPUT <code>"Alice"</code>

Q13 Title-case a full name

EASY

.title()

Convert a full name entered in lowercase to title case.

INPUT

name = "john michael doe"

EXPECTED OUTPUT

"John Michael Doe"

Q14 Check if a string contains only digits

EASY

.isdigit()

A user entered their age. Check if the input contains only numeric digits.

INPUT

s1 = "2004"
s2 = "20a4"

EXPECTED OUTPUT

True
False

Q15 Check if a string is alphabets only

EASY

.isalpha()

Validate that a name field contains only letters and no numbers or symbols.

INPUT

s1 = "Alice"
s2 = "Ali3e"

EXPECTED OUTPUT

True
False

Q16 Strip only left-side whitespace

EASY

.lstrip()

Remove leading spaces from the string but keep trailing spaces intact.

INPUT

s = " Hello "

EXPECTED OUTPUT

"Hello "

Q17 Strip only right-side whitespace

EASY	<code>.rstrip()</code>
Remove trailing spaces from the string but keep leading spaces intact.	
INPUT s = " Hello "	EXPECTED OUTPUT " Hello"

Q18 Check if all characters are uppercase	
EASY	<code>.isupper()</code>
Check if the given string is entirely in uppercase letters.	
INPUT s1 = "PYTHON" s2 = "Python"	EXPECTED OUTPUT True False

Q19 Check if all characters are lowercase	
EASY	<code>.islower()</code>
Check whether the string is entirely in lowercase.	
INPUT s1 = "hello" s2 = "Hello"	EXPECTED OUTPUT True False

Q20 Centre-align a string within a width	
EASY	<code>.center()</code>
Centre the word 'Python' within a total width of 20 characters, padded with dashes.	
INPUT s = "Python"	EXPECTED OUTPUT "-----Python-----"

Section 3 — Intermediate

These 10 questions combine multiple string methods or apply them in realistic scenarios like formatting reports, validating inputs, and parsing data.

Q21 Swap the case of every character

INTERMEDIATE

`.swapcase()`

Given a mixed-case string, convert uppercase to lowercase and vice versa.

INPUT

```
s = "Hello World"
```

EXPECTED OUTPUT

```
"hELLO wORLD"
```

Q22 Check if a string is alphanumeric

INTERMEDIATE

`.isalnum()`

Validate a username that should contain only letters and digits, no spaces or symbols.

INPUT

```
s1 = "User123"
```

```
s2 = "User 123"
```

EXPECTED OUTPUT

```
True
```

```
False
```

Q23 Left-pad a number with zeros

INTERMEDIATE

`.zfill()`

A student ID must always be 6 digits wide. Pad with leading zeros if shorter.

INPUT

```
id = "42"
```

EXPECTED OUTPUT

```
"000042"
```

Q24 Split a CSV line into fields

INTERMEDIATE

`.split(',')`

Given a CSV string, split it by commas to extract individual values.

INPUT

```
row = "Alice,25,Math,88"
```

EXPECTED OUTPUT

```
['Alice', '25', 'Math', '88']
```

Q25 Right-align a string within a width

INTERMEDIATE

`.rjust()`

Right-align a product name within a width of 20 characters for a formatted receipt.

INPUT s = "Apple"	EXPECTED OUTPUT "Apple"
-----------------------------	-----------------------------------

Q26 Left-align a string within a width	
INTERMEDIATE	.ljust()
Left-align a student name within 15 characters so scores line up neatly.	
INPUT s = "Alice"	EXPECTED OUTPUT "Alice"

Q27 Check if a string contains only spaces	
INTERMEDIATE	.isspace()
After stripping input, check whether the remaining content is purely whitespace.	
INPUT s1 = " " s2 = " a "	EXPECTED OUTPUT True False

Q28 Format a greeting using f-string	
INTERMEDIATE	f-string + .upper()
Given a name and a city, build a formatted greeting where the city appears in uppercase.	
INPUT name = "Sahil" city = "toronto"	EXPECTED OUTPUT "Welcome, Sahil! You are from TORONTO."

Q29 Extract domain from an email address	
INTERMEDIATE	.split() + indexing
Given an email address, extract only the domain part (everything after the @).	
INPUT email = "sahil@example.com"	EXPECTED OUTPUT "example.com"

Q30 Build a formatted student report line	
INTERMEDIATE	<code>.ljust() + .rjust() + str()</code>
Format a student record as a fixed-width line: name left-aligned in 15 chars, score right-aligned in 5 chars.	
INPUT name="Alice", score=88, grade="B"	EXPECTED OUTPUT "Alice 88 B"

Section 4 — LeetCode-Style Challenges

All 6 questions are solvable using only string methods — no loops, no if/else, no imports. Students must think creatively with the tools they already know.

QLC1 Is it a palindrome?	
EASY	<code>.lower() · .replace() · slicing</code>
Given a string, determine whether it reads the same forwards and backwards. Ignore spaces and letter case. Return True or False.	
INPUT s = "Race car" s = "hello" s = "A man a plan a canal Panama"	EXPECTED OUTPUT True False True

QLC2 Count the vowels	
EASY	<code>.lower() · .count()</code>
Given a sentence, count the total number of vowels (a, e, i, o, u) it contains. Ignore case.	
INPUT s = "Hello World" s = "Python"	EXPECTED OUTPUT 3 1

QLC3 Reverse the words in a sentence	
EASY	<code>.split() · .join() · slicing</code>

Given a sentence, reverse the order of the words. Individual letters within each word stay the same.

INPUT

```
s = "Hello World"
s = "I love Python"
```

EXPECTED OUTPUT

```
"World Hello"
"Python love I"
```

QLC4 Mask a credit card number**MEDIUM**

`.replace()` · `slicing` · `.join()`

Given a 16-digit credit card number (with or without dashes), return a masked version where all digits except the last 4 are replaced with *. Format in groups of 4 separated by dashes.

INPUT

```
card = "1234-5678-9012-3456"
card = "1234567890123456"
```

EXPECTED OUTPUT

```
"*****-****-3456"
"*****-****-3456"
```

QLC5 Find the most frequent character**MEDIUM**

`.lower()` · `.replace()` · `.count()` · `max()`

Given a string, find the letter that appears most often. Ignore case and spaces. On a tie, return the one that appears first.

INPUT

```
s = "hello world"
s = "banana"
```

EXPECTED OUTPUT

```
"l" (3 times)
"a" (3 times)
```

QLC6 Build a username from a full name**MEDIUM**

`.lower()` · `.split()` · `.strip()` · `slicing`

Given a person's full name, generate a username: take the first letter of the first name, add the full last name, convert to lowercase. Skip the middle name if present.

INPUT

```
name = "John Michael Smith"
name = "Alice Johnson"
```

EXPECTED OUTPUT

```
"jsmith"
"ajohnson"
```

